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JupyterLab Python [conda env:base] * Anaconda Toolbox

[2]: import pandas as pd
df=pd.read_excel("Supermarket-Sales-Sample-Data.xlsx")
print(df)

0 NaN Excel Sample Data NaN
1 NaN NaN NaN
2 NaN Supermarket Sales Data NaN
3 NaN NaN NaN
4 NaN NaN NaN
.. ..
72 NaN 1066 2024-02-02 00:00:00
73 NaN 1067 2024-02-03 00:00:00
74 NaN 1068 2024-02-04 00:00:00
75 NaN 1069 2024-02-04 00:00:00
76 NaN 1070 2024-02-04 00:00:00

0 NaN NaN NaN NaN NaN
1 NaN NaN NaN NaN NaN
2 NaN NaN NaN NaN NaN
3 NaN NaN NaN NaN Tax
4 NaN NaN NaN NaN
.. ..
72 Sarah Gonzalez 2024-02-06 00:00:00 79.99 2 15.998
73 Matthew Smith 2024-02-06 00:00:00 49.99 3 14.997
74 Emily Johnson 2024-02-05 00:00:00 129.99 1 12.999
75 Daniel Brown 2024-02-08 00:00:00 19.99 4 7.996
76 Jennifer Hernandez 2024-02-07 00:00:00 149.99 1 14.999

Unnamed: 8

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```
[2]: import pandas as pd
df=pd.read_excel("Supermarket-Sales-Sample-Data.xlsx")
print(df)
```

	Unnamed: 0	Unnamed: 1	Unnamed: 2	\
0	NaN	Excel Sample Data	NaN	
1	NaN	NaN	NaN	
2	NaN	Supermarket Sales Data	NaN	
3	NaN	NaN	NaN	
4	NaN	NaN	NaN	
..	...	...	...	
72	NaN	1066	2024-02-02 00:00:00	
73	NaN	1067	2024-02-03 00:00:00	
74	NaN	1068	2024-02-04 00:00:00	
75	NaN	1069	2024-02-04 00:00:00	
76	NaN	1070	2024-02-04 00:00:00	

	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	\
0	NaN	NaN	NaN	NaN	NaN	
1	NaN	NaN	NaN	NaN	NaN	
2	NaN	NaN	NaN	NaN	NaN	
3	NaN	NaN	NaN	NaN	NaN	
4	NaN	NaN	NaN	NaN	Tax	
..	...	...	...	...	...	
72	Sarah Gonzalez	2024-02-06 00:00:00	79.99	2	15.998	
73	Matthew Smith	2024-02-06 00:00:00	49.99	3	14.997	
74	Emily Johnson	2024-02-05 00:00:00	129.99	1	12.999	
75	Daniel Brown	2024-02-08 00:00:00	19.99	4	7.996	
76	Jennifer Hernandez	2024-02-07 00:00:00	149.99	1	14.999	

	Unnamed: 8	...
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[2]:

```
import pandas as pd
df=pd.read_excel("Supermarket-Sales-Sample-Data.xlsx")
print(df)
```

	Unnamed: 0	Unnamed: 1	Unnamed: 2	\
0	NaN	Excel Sample Data	NaN	
1	NaN	NaN	NaN	
2	NaN	Supermarket Sales Data	NaN	
3	NaN	NaN	NaN	
4	NaN	NaN	NaN	
..	...	...	...	
72	NaN	1066	2024-02-02 00:00:00	
73	NaN	1067	2024-02-03 00:00:00	
74	NaN	1068	2024-02-04 00:00:00	
75	NaN	1069	2024-02-04 00:00:00	
76	NaN	1070	2024-02-04 00:00:00	

	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	\
0	NaN	NaN	NaN	NaN	NaN	
1	NaN	NaN	NaN	NaN	NaN	
2	NaN	NaN	NaN	NaN	NaN	
3	NaN	NaN	NaN	NaN	NaN	
4	NaN	NaN	NaN	NaN	Tax	
..	...	...	...	...	...	
72	Sarah Gonzalez	2024-02-06 00:00:00	79.99	2	15.998	
73	Matthew Smith	2024-02-06 00:00:00	49.99	3	14.997	
74	Emily Johnson	2024-02-05 00:00:00	129.99	1	12.999	
75	Daniel Brown	2024-02-08 00:00:00	19.99	4	7.996	
76	Jennifer Hernandez	2024-02-07 00:00:00	149.99	1	14.999	

	Unnamed: 8	...
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JupyterLab Python [conda env:base] * Anaconda Toolbox

```
[3]: import pandas as pd
df=pd.read_excel("Supermarket-Sales-Sample-Data.xlsx")
df.head(100)
```

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnamed: 8
0	NaN	Excel Sample Data	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	Supermarket Sales Data	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	Tax	0.1
...	...	...	...	...	...	...	...	...	...
72	NaN	1066	2024-02-02 00:00:00	Sarah Gonzalez	2024-02-06 00:00:00				
73	NaN	1067	2024-02-03 00:00:00	Matthew Smith	2024-02-06 00:00:00				
74	NaN	1068	2024-02-04 00:00:00	Emily Johnson	2024-02-05 00:00:00				
75	NaN	1069	2024-02-04 00:00:00	Daniel Brown	2024-02-08 00:00:00				
76	NaN	1070	2024-02-04 00:00:00	Jennifer Hernandez	2024-02-07 00:00:00				

77 rows x 9 columns

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Python [conda env:base] * Anaconda Toolbox

77 rows x 9 columns

[4]: import pandas as pd
df=pd.read_csv("people-100.csv")
print(df)

	Index	User Id	First Name	Last Name	Sex	
0	1	88F7B33d2bcf9f5	Shelby	Terrell	Male	
1	2	f90cD3E76f1A9b9	Phillip	Summers	Female	
2	3	DbeAb8CcdfeFC2c	Kristine	Travis	Male	
3	4	A31Bee3c201ef58	Yesenia	Martinez	Male	
4	5	1bA7A3dc874da3c	Lori	Todd	Male	
..	...	...	...	...	...	
95	96	5eFda7caAeB260E	Dennis	Barnes	Female	
96	97	CCbFce93d3720bE	Steve	Patterson	Female	
97	98	2fEc528aFAF0b69	Wesley	Bray	Male	
98	99	Adc7ad9B6e4A1Fe	Summer	Oconnell	Female	
99	100	b8D0aD3490FC7e1	Mariah	Bernard	Male	

	Email	Phone	Date of birth	
0	elijah57@example.net	001-084-906-7849x73518	1945-10-26	
1	bethany14@example.com	214.112.6044x4913	1910-03-24	
2	bthompson@example.com	277.609.7938	1992-07-02	
3	kaitlinkaiser@example.com	584.094.6111	2017-08-03	
4	buchananmanuel@example.net	689-207-3558x7233	1938-12-01	
..	...	...	...	
95	bmartin@example.org	001-095-524-2112x257	1954-07-30	
96	latasha46@example.net	001-865-478-5157	1932-04-29	
97	regina11@example.org	995-542-3004x76800	1994-12-28	
98	alexiscantrell@example.org	001-273-685-6932x092	2012-04-12	
99	pcopeland@example.org	(341)594-6554x44657	2016-11-15	

Job Title

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Code

```
[6]: import pandas as pd
df=pd.read_json("titanics.json")
print(df)
df.head()
```

	PassengerId	Survived	Pclass	
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	
..	...	...	...	
886	887	0	2	
887	888	1	1	
888	889	0	3	
889	890	1	1	
890	891	0	3	

	Name	Sex	Age	SibSp	
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	
..	...	...	...	...	
886	Montvila, Rev. Juozas	male	27.0	0	
887	Graham, Miss. Margaret Edith	female	19.0	0	
888	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	
889	Behr, Mr. Karl Howell	male	26.0	0	
890	Dooley, Mr. Patrick	male	32.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	None	S
1	0	PC 17599	71.2833	C85	C

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[7]: #Check missing values

[8]: #is null
import pandas as pd
df=pd.read_json("titanics.json")
print(df.isnull())

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	\
0	False	False	False	False	False	False	False	False	False	
1	False	False	False	False	False	False	False	False	False	
2	False	False	False	False	False	False	False	False	False	
3	False	False	False	False	False	False	False	False	False	
4	False	False	False	False	False	False	False	False	False	
..	...	...	...	...	...	...	...	...	...	
886	False	False	False	False	False	False	False	False	False	
887	False	False	False	False	False	False	False	False	False	
888	False	False	False	False	False	True	False	False	False	
889	False	False	False	False	False	False	False	False	False	
890	False	False	False	False	False	False	False	False	False	
	Fare	Cabin	Embarked							
0	False	True	False							
1	False	False	False							
2	False	True	False							
3	False	False	False							
4	False	True	False							
..	...	...	...							
886	False	True	False							
887	False	False	False							
888	False	True	False							
889	False	False	False							
890	False	True	False							

[891 rows x 12 columns]

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JupyterLab Python [conda env:base] * Anaconda Toolbox

```
889 False False False
890 False True False

[891 rows x 12 columns]
```

[9]: *#count missing values for all fields*

```
import pandas as pd
df=pd.read_json("titanics.json")
print(df.isnull().sum())
```

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2
dtype:	int64

[10]: *#Find rows having missing*

```
import pandas as pd
df=pd.read_json("titanics.json")
print(df)
df[df['Age'].isnull()]
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

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```
dtype: int64

[10]: #Find rows having missing
import pandas as pd
df=pd.read_json("titanics.json")
print(df)
df[df['Age'].isnull()]
```

	PassengerId	Survived	Pclass	
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	
..	...	...	...	
886	887	0	2	
887	888	1	1	
888	889	0	3	
889	890	1	1	
890	891	0	3	

	Name	Sex	Age	SibSp	
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	
..	...	...	...	...	
886	Montvila, Rev. Juozas	male	27.0	0	
887	Graham, Miss. Margaret Edith	female	19.0	0	
888	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	
889	Behr, Mr. Karl Howell	male	26.0	0	
890	Dooley, Mr. Patrick	male	32.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	None	S

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[12]: #by using mode() means the most frequently occurring value in a column

[13]:

```
import pandas as pd
df=pd.read_json("titanics.json")
print(df)
mode_embarked=df['Embarked'].mode()[0]
df['Embarked']=df['Embarked'].fillna(mode_embarked)
#drop remove missing data
df.dropna()
#remove column
df.dropna(axis=1)
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	
..	...	...	...	
886	887	0	2	
887	888	1	1	
888	889	0	3	
889	890	1	1	
890	891	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	
..	...	...	...	...	
886	Montvila, Rev. Juozas	male	27.0	0	
887	Graham, Miss. Margaret Edith	female	19.0	0	

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891 rows x 10 columns

[14]: #for specific row and column

```
[15]: import pandas as pd
df=pd.read_json("titanics.json")
print(df.iloc[0:5,1:3])
```

	Survived	Pclass
0	0	3
1	1	1
2	1	3
3	1	1
4	0	3

[16]: #change index to customer id

```
[22]: import pandas as pd
df=pd.read_csv("customers-100.csv")
print(df)
```

	Index	Customer Id	First Name	Last Name
0	1	DD37Cf93aecA6Dc	Sheryl	Baxter
1	2	1Ef7b82A4CAAD10	Preston	Lozano
2	3	6F94879bDAfE5a6	Roy	Berry
3	4	5Cef8BFA16c5e3c	Linda	Olsen
4	5	053d585Ab6b3159	Joanna	Bender
..	...	...	...	...
95	96	cb8E23e48d22Eae	Karl	Greer
96	97	CeD220bdAacfaDf	Lynn	Atkinson
97	98	28CDbC0dFe4b1Db	Fred	Guerra

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[20]: import pandas as pd
df=pd.read_csv("customers-100.csv",index_col='Customer Id')
print(df)

	Index	First Name	Last Name	Company \
Customer Id				
DD37Cf93aecA6Dc	1	Sheryl	Baxter	Rasmussen Group
1Ef7b82A4CAAD10	2	Preston	Lozano	Vega-Gentry
6F94879bDAfE5a6	3	Roy	Berry	Murillo-Perry
5Cef8BFA16c5e3c	4	Linda	Olsen	Dominguez, Mcmillan and Donovan
053d585Ab6b3159	5	Joanna	Bender	Martin, Lang and Andrade
...	...	...	...	...
cb8E23e48d22Eae	96	Karl	Greer	Carey LLC
CeD220bdAaCfaDf	97	Lynn	Atkinson	Ware, Burns and Oneal
28CDbC0dFe4b1Db	98	Fred	Guerra	Schmitt-Jones
c23d1D9EE8DEB0A	99	Yvonne	Farmer	Fitzgerald-Harrell
2354a0E336A91A1	100	Clarence	Haynes	Le, Nash and Cross

	City	Country \
Customer Id		
DD37Cf93aecA6Dc	East Leonard	Chile
1Ef7b82A4CAAD10	East Jimmychester	Djibouti
6F94879bDAfE5a6	Isabelborough	Antigua and Barbuda
5Cef8BFA16c5e3c	Bensonview	Dominican Republic
053d585Ab6b3159	West Priscilla	Slovakia (Slovak Republic)
...	...	...
cb8E23e48d22Eae	East Richard	Guyana
CeD220bdAaCfaDf	New Bradview	Sri Lanka
28CDbC0dFe4b1Db	Ortegaland	Solomon Islands
c23d1D9EE8DEB0A	Lake Elijahview	Aruba
2354a0E336A91A1	Judymouth	Honduras

	Phone 1	Phone 2 \
Customer Id		
DD37Cf93aecA6Dc	330-637-5454	303-884-8510-740

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[100 rows x 11 columns]

[21]: #accessing specific columns for particular index

```
[41]: import pandas as pd
df=pd.read_csv("customers-100.csv",index_col='Customer Id')
print(df.loc['053d585Ab6b3159',['First Name','Company']])
print(df.loc['1Ef7b82A4CAAD10':'053d585Ab6b3159',['First Name','Company']])
```

First Name	Joanna
Company	Martin, Lang and Andrade
Name: 053d585Ab6b3159, dtype: object	
First Name	Company
Customer Id	
1Ef7b82A4CAAD10	Preston Vega-Gentry
6F94879bDAfE5a6	Roy Murillo-Perry
5Cef88FA16c5e3c	Linda Dominguez, Mcmillan and Donovan
053d585Ab6b3159	Joanna Martin, Lang and Andrade

[31]: #groupby

```
[39]: import pandas as pd
df=pd.read_json("titanics.json")
print(df.groupby('Pclass')['Fare'].mean())
df.groupby('Sex')['Survived'].sum()
print(df.groupby('Pclass').agg({
    'Age':['mean','min','max'],
    'Fare':['mean','sum']
})))
#transfirm
#returns same number of rows
df['Avg_Class_Fare']=df.groupby('Pclass')['Fare'].transform('mean')
print(df)
print(df.columns)
```

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```
[31]: #groupby

[39]: import pandas as pd
df=pd.read_json("titanics.json")
print(df.groupby('Pclass')['Fare'].mean())
df.groupby('Sex')['Survived'].sum()
print(df.groupby('Pclass').agg({
    'Age':['mean','min','max'],
    'Fare':['mean','sum']
}))
#transfirm
#returns same number of rows
df['Avg_Class_Fare']=df.groupby('Pclass')['Fare'].transform('mean')
print(df)
print(df.columns)
d

Pclass
1    84.154687
2    20.662183
3    13.675550
Name: Fare, dtype: float64

      Age      Fare
      mean  min  max  mean  sum
Pclass
1    38.233441  0.92  80.0  84.154687  18177.4125
```

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Code

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```
Age
0 20.0
1 25.0
2 30.0
3 35.0
4 40.0
```

[40]: `#interpolate`
`sample=pd.DataFrame({`
 `'Age': [20,25,None,35,40]`
`})`
`sample['Age']=sample['Age'].interpolate()`
`print(sample)`

```
Age
0 20.0
1 25.0
2 30.0
3 35.0
4 40.0
```

[47]: `import pandas as pd`
`students=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="students")`
`marks=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="marks")`
`contact=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="contact")`

#1. INNER MERGE
#returns only matching Student_IDs from both DataFrames.

```
inner_df=pd.merge(
    students,
    marks,
    on="Student_ID",
    how="inner"
```

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```
1 25.0
2 30.0
3 35.0
4 40.0

[47]: import pandas as pd
students=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="students")
marks=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="marks")
contact=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="contact")
```

```
#1. INNER MERGE
#returns only matching Student_IDs from both DataFrames.
```

```
inner_df=pd.merge(
    students,
    marks,
    on="Student_ID",
    how="inner"
)
print(inner_df.head())
print(inner_df.shape)
```

	Student_ID	Name	Department	Maths	Python
0	1001	Student_1	CSE	61	56
1	1002	Student_2	IT	62	57
2	1003	Student_3	ECE	63	58
3	1004	Student_4	EEE	64	59
4	1005	Student_5	MECH	65	60

```
[52]: #Left Merge
#kept all records from the Left DataFrame
```

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[52]:
#Left Merge
#kept all records from the Left DataFrame

import pandas as pd
students=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="students")
marks=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="marks")
contact=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="contact")

left_df=pd.merge(
 students,
 marks,
 on="Student_ID",
 how="left"
)
print(left_df)

	Student_ID	Name	Department	Maths	Python
0	1001	Student_1	CSE	61.0	56.0
1	1002	Student_2	IT	62.0	57.0
2	1003	Student_3	ECE	63.0	58.0
3	1004	Student_4	EEE	64.0	59.0
4	1005	Student_5	MECH	65.0	60.0
5	1006	Student_6	CSE	66.0	61.0
6	1007	Student_7	IT	67.0	62.0
7	1008	Student_8	ECE	68.0	63.0
8	1009	Student_9	EEE	69.0	64.0
9	1010	Student_10	MECH	70.0	65.0
10	1011	Student_11	CSE	71.0	66.0
11	1012	Student_12	IT	72.0	67.0
12	1013	Student_13	ECE	73.0	68.0
13	1014	Student_14	EEE	74.0	69.0
14	1015	Student_15	MECH	75.0	70.0
15	1016	Student_16	CSE	76.0	71.0

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44	1045	Student_45	MECH	65	60
45	1046	Student_46	CSE	NaN	NaN
46	1047	Student_47	IT	NaN	NaN
47	1048	Student_48	ECE	NaN	NaN
48	1049	Student_49	EEE	NaN	NaN
49	1050	Student_50	MECH	NaN	NaN

```
[54]: import pandas as pd
students=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="students")
marks=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="marks")
contact=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="contact")

#Right Merge
right_df=pd.merge(
    students,
    marks,
    on="Student_ID",
    how="right"
)
print(right_df)
print(right_df.tail())
```

	Student_ID	Name	Department	Maths	Python
0	1001	Student_1	CSE	61	56
1	1002	Student_2	IT	62	57
2	1003	Student_3	ECE	63	58
3	1004	Student_4	EEE	64	59
4	1005	Student_5	MECH	65	60
5	1006	Student_6	CSE	66	61
6	1007	Student_7	IT	67	62
7	1008	Student_8	ECE	68	63
8	1009	Student_9	EEE	69	64
9	1010	Student_10	MECH	70	65

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```
[55]: outer_df=pd.merge(
      students,
      marks,
      on="Student_ID",
      how="outer"
    )
      print(outer_df.head())
      print(outer_df.shape)
```

	Student_ID	Name	Department	Maths	Python
0	1001	Student_1	CSE	61.0	56.0
1	1002	Student_2	IT	62.0	57.0
2	1003	Student_3	ECE	63.0	58.0
3	1004	Student_4	EEE	64.0	59.0
4	1005	Student_5	MECH	65.0	60.0

```
(55, 5)
```

```
[56]: #Merge Multiple Tables
```

```
[60]: import pandas as pd
      students=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="students")
      marks=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="marks")
      contact=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="contact")
      student_complete=pd.merge(
        students,
        marks,
        on="Student_ID",
        how="right"
      )
      student_complete=pd.merge(
        students,
```

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3	1004	Student_4	EEE	64.0	59.0
4	1005	Student_5	MECH	65.0	60.0

(55, 5)

```
[56]: #Merge Multiple Tables

[60]: import pandas as pd
students=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="students")
marks=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="marks")
contact=pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx",sheet_name="contact")
student_complete=pd.merge(
    students,
    marks,
    on="Student_ID",
    how="right"
)
student_complete=pd.merge(
    students,
    contact,
    on="Student_ID",
    how="right"
)

print(student_complete.tail())
```

	Student_ID	Name	Department	Email
45	1046	Student_46	CSE	student46@college.edu
46	1047	Student_47	IT	student47@college.edu

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